

Take-Home: Build a Voice Ordering Agent

A scoped proof-of-concept to design, build, and reason about a real-time AI voice agent — a backend the model drives through tools, plus frontend to talk to it.

1. The scenario

A restaurant wants an assistant that answers the phone and takes orders. Build a working POC: a caller can **ask what's on the menu**, **order an item** (including options like size, toppings, or a side), **hear it confirmed**, and **place it**. And send confirmation. Keep the menu comprehensive including nested modifiers.

caller speaks	agent (LLM)	picks a tool (list_menu / add_to_cart / ...)
	updates the order	replies UI shows transcript + live cart and order

2. Build this

Backend

- A comprehensive **menu** with nested modifiers. At least one item has a **required** choice — e.g. a pizza needs a size, or a burrito needs a protein. Bonus: one **nested** required choice (a combo → pick an entrée → pick that entrée's side).
- An **LLM agent that drives the order through tools**. The model decides what to say and which tool to call. It must **never invent** items or prices.
- A **session per call** — its own cart and state (two callers ≠ one shared cart).
- An endpoint your frontend connects to.
- On Order Confirmation send **SMS** and **E-mail** via tool call.
- Make sure to handle **Call Termination** and cover all the necessary cases.

Frontend

- A web page to talk to the agent. Should be seamless as if speaking to a real person.

- **Live transcript** — both caller and agent lines. Including tokens consumption and latency
- A **live order panel** that updates as items are added / changed.
- Add a **Orders page** which shows whether the orders are placed successfully once the agent acknowledges

3. Core vs. stretch

Do the core well first. Then pick stretch items that show **your** strengths — nobody finishes them all, and two done cleanly beat six half-built.

Core — we expect these working

- Text agent with real **tool-calling**
- Menu Q&A answered **from the tools** (no hallucination)
- Add an item **with a required option**, view cart, place order
- Frontend showing **transcript + live cart**

Make sure to include

- **Real voice** — speech-to-speech, or STT LLM TTS
- **Latency** — measure & show time-to-first-response; stream replies
- **Interruptions / barge-in**
- **Nested modifiers** — item option sub-option, all enforced
- Context management for long conversations
- Tests — unit, or an automated scripted conversation
- A tiny **analytics** view (turns, latency, tokens, cost)
- One-command run (Docker / compose)

4. Rules

- **Any language / framework** — but tell us *why* in your README.
- **Any LLM provider**

5. What to hand in

- A **git repo** (or zip) with a **README**: how to run it · key decisions & tradeoffs · what you'd do next with more time.
- A **2-3 minute screen recording** (or a short live walkthrough) showing a full order end-to-end.
